RAGHAVENDRA SHARMA

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SUMMARY

Electrical and Computer Engineering student at Virginia Tech with hands-on experience in AI/ML systems, computer vision, and software development. Proven ability to design production-grade pipelines achieving 99.28% accuracy, lead student communities, and deliver results across multidisciplinary projects. Actively seeking internship opportunities to apply expertise in AI, embedded systems, and full-stack development.

EDUCATION

Virginia Tech | Blacksburg, VA *Expected May 2028*

B.S. in Electrical and Computer Engineering

Minors in Quantum Computing, Computer Science, and Mathematics (in progress)

EXPERIENCE

AWR Group | Dubai, UAE (Hybrid) *May 2026 – Jul 2026*

AI & Computer Vision Intern

- **99.28% VIN extraction accuracy**o-end AI document-understanding system for extracting vehicle information from UAE Mulkiya (vehicle registration) cards, achieving
- Built a custom dataset from scratch — collected, cleaned, deduplicated, and validated thousands of vehicle registration images; developed automated auditing tools to flag missing images, invalid VINs, and annotation inconsistencies.
- Benchmarked four OCR engines (EasyOCR, PaddleOCR, RapidOCR, OpenCV) within a custom evaluation framework that generated accuracy reports and tracked per-engine failure modes.
- Engineered a hybrid AI pipeline combining rule-based VIN validation, edit-distance correction, candidate ranking, and Vision Language Model (Qwen2.5-VL) fallback; dynamic confidence-based routing cut expensive model inference calls significantly.
- Integrated Qwen2.5-VL with custom prompt engineering for structured JSON field extraction, chassis-number localization, and multi-stage fallback strategies on difficult images.
- Resolved complex dependency conflicts (PyTorch, Transformers, PaddleOCR, CUDA, NumPy, OpenCV) and optimized GPU inference pipelines for large-scale document processing.
- Technologies: Python, PyTorch, Transformers, Qwen2.5-VL, EasyOCR, PaddleOCR, RapidOCR, OpenCV, Pandas, NumPy, Hugging Face Ecosystem, Git.

Hybrid Electric Vehicle Team (HEVT), Virginia Tech Sep 2025 - Present

Electrical / Integration Team Member

- Contribute to HEVT's electrical/integration subteam, supporting multidisciplinary electric-vehicle development.
- Focus areas: electrical hardware integration, circuit analysis, embedded systems, and hardware-software interfaces.

Competitive Robotics Organization (CRO), Virginia Tech Aug 2025 - May 2026

VEX U Electrical & Programming Team Member

- Contributed to electrical development and robot programming within CRO's VEX U competition team.
- Emphasis on robot electrical systems and hardware-software integration alongside competition-robot programming.

Cybersecurity Club, Virginia Tech | Blacksburg, VA *Jan 2025 – May 2025*

Vice President

- Led a 45-member student organization; organized CTF competitions, ethical-hacking workshops, and industry guest-speaker events, growing club participation by 8%.
- Mentored members in Kali Linux, AI-driven security trends, and cybersecurity tooling; managed budgets, sponsor relationships, and event logistics.

Virginia Tech | Blacksburg, VA *Aug 2025 – Present*

Inclusion & Belonging Student Leader

- Lead a residential community of 130 students; grew event participation by 12% through weekly micro-events and inclusive programming.
- Conducted proactive 1:1 check-ins, triaged student concerns, and facilitated warm handoffs to Hokie Wellness, Cook Counseling, and the Dean of Students office.

AMITY University Dubai | Dubai, UAE *May 2024 – Jul 2024*

Design Intern

- Designed and fabricated a functioning small-scale windmill prototype using CAD/3D modeling, engineering design principles, and hands-on fabrication tools (drills, saws, soldering).
- Tested and optimized the prototype for stable energy generation; donated the completed unit to a community organization as a sustainable energy demonstration tool.

Nanyang Technological University & Hewlett Packard Enterprise | Remote *Jun 2023 – Sep 2023*

AI Research Intern

- Assisted researchers on an AI project — prepared datasets, cleaned labels, ran experiments, and documented results within a cross-functional team of 7 students.
- Technologies: Python, scikit-learn, TensorFlow, PyTorch, Pandas.

Indian Space Research Organisation (ISRO) | Remote *Mar 2023 – Jun 2023*

Rocket Design Intern

- Contributed to preliminary rocket design tasks: CAD sketches, basic structural calculations, requirements tracking, and safety protocols; produced a 3D rocket model and summary report.

PROJECTS

Collision Repair Cost Estimator | *Python, Keras, TensorFlow, ResNet50, InceptionResNetV2, NumPy, Pandas, Matplotlib*

- Built a transfer-learning classifier to label vehicle damage (minor / moderate / severe) on a 1,400-image dataset, reaching ~71% accuracy; benchmarked ResNet50 vs. InceptionResNetV2 and proposed repair-cost regression extension.

github.com/RaghavendraSharma28/AIM

Car Price Prediction | *R, HTML, CSS, JavaScript*

- Trained a regression model on the Kaggle Car Price Prediction dataset and built a prototype front-end UI; currently integrating the front-end with the R backend for seamless local execution.

github.com/RaghavendraSharma28/CarPricePredictionUsingR

WB Studios | Remote Aug 2024 - Present

Founder & Lead Developer

- Designed, built, and published Roblox games that have reached 4.5+ million cumulative plays, writing the Lua, designing the UI, and running testing and iterative releases.
- Grew the player community to 64,000+ members while managing the full product lifecycle from concept through release, updates, and player feedback.
- Led a distributed contributor team using Agile-style task planning, version control, and shared development practices.
- Brought AI-assisted development into the studio through MCP, building custom tools and skills for contributors, and contributed to an open-source MCP server.
- Shared voxel destruction and aircraft flight physics techniques with the Roblox developer community.
- Negotiated with investors and sold ownership stakes in two games.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, Assembly, R, SQL, Lua, JavaScript, HTML/CSS; Bash (basic)

FPGA & Digital Design: Verilog, SystemVerilog, RTL design, finite-state machines, testbenches; Quartus Prime, Questa; DE1-SoC, DE10-Lite

Embedded Systems: MSP432P401R, ARM Cortex-M, embedded C/C++, GPIO, button/LED and LCD interfacing; Code Composer Studio

Circuits & Hardware: PCB design, LTspice, MATLAB, Analog Discovery 2/3, WaveForms, oscilloscope/DMM measurements, soldering; CAD/3D modeling, SolidWorks, fabrication, robotics, electrical integration, robot programming

AI / ML & Vision: PyTorch, TensorFlow, Keras, scikit-learn, Transformers, Hugging Face, Qwen2.5-VL, VLMs, prompt engineering; OpenCV, EasyOCR, PaddleOCR, RapidOCR, image preprocessing, document AI

Data Engineering: Pandas, NumPy, Matplotlib, dataset curation, annotation validation, evaluation pipelines

Software & DevOps: Git, Docker, Linux, CMake/CTest, VS Code, CLI tooling, AWS, GPU/CUDA; React/Vite, FastAPI, RAG, MCP

Collaboration: Technical writing, project management, team leadership, mentoring, Agile task planning

AWARDS

Academic Excellence in Physics | GEMS International School, May 2023

- Recognized for outstanding academic performance in Physics, demonstrating strong analytical and problem-solving abilities.